

# TZ-YING (GINA) WU

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## WORK EXPERIENCE

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- **Staff AI Research Scientist, Intel Multimodal AI** *Present*
  - Leading applied research projects on multimodal and agentic AI systems for enterprise applications.
- **AI Research Scientist, Intel AI Lab** *Aug. 2024 - Apr. 2026*
  - Conducted research on video understanding and generation, building VLMs and multimodal LLMs for cross-modal reasoning and inference-time scaling techniques for video diffusion models.
  - Collaborated cross-functionally to deploy AI systems for video searching and summarization. *[Link]*
- **Machine Learning Research Intern, Intel AI Lab** *Jun. 2023 - Mar. 2024*
  - Proposed a lightweight adaptation method for video foundation models, achieving high fine-tuning performance by optimizing  $< 1\%$  of model parameters. Work published at WACV 2025. *[Paper]*
- **Applied Scientist Intern, Amazon AWS AI Lab** *Jun. 2021 - Sep. 2021*
  - Developed an efficient solution for large-scale class-incremental learning by addressing the stability-plasticity dilemma and mitigating data imbalance. Work published at CVPR 2022. *[Paper]*

## EDUCATION

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- **University of California San Diego (UCSD), La Jolla, CA** *Sep. 2018 - Jun. 2024*
  - Ph.D. in Electrical and Computer Engineering (advised by Prof. Nuno Vasconcelos)
- **National Tsing Hua University (NTHU), Taiwan** *Sep. 2012 - Jan. 2018*
  - Master of Science in Electrical Engineering (advised by Prof. Min Sun)
  - Bachelor of Science in Electrical Engineering; GPA: 4.02/4.3 (graduated a half-year earlier)

## SELECTED PUBLICATIONS

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- “Search2Motion: Training-Free Object-Level Motion Control via Attention-Consensus Search” (*Preprint*)
- Shubhashis Roy Dipta\*, **Tz-Ying Wu\***, Subarna Tripathi, “VC-Inspector: Advancing Reference-free Evaluation of Video Captions with Factual Analysis”, In ACL (main), 2026. *[Paper]*
- **Tz-Ying Wu**, Sharath Nittur Sridhar, Subarna Tripathi, “Harnessing Object Grounding for Time-Sensitive Video Understanding”, In WACV, 2026. *[Paper]*
- Ivan Rodin\*, **Tz-Ying Wu\***, Kyle Min, Sharath Nittur Sridhar, Antonino Furnari, Subarna Tripathi, Giovanni Maria Farinella, “EASG-Bench: Video Q&A Benchmark with Egocentric Action Scene Graphs”, In ICCVW, 2025. *[Paper]*
- **Tz-Ying Wu**, Kyle Min, Subarna Tripathi, Nuno Vasconcelos, “Ego-VPA: Egocentric Video Understanding with Parameter-efficient Adaptation”, In WACV, 2025. *[Paper]*
- **Tz-Ying Wu\***, Chih-Hui Ho\*, and Nuno Vasconcelos, “ProTeCt: Prompt Tuning for Taxonomic Open Set Classification”, In CVPR, 2024. *[Paper]*
- **Tz-Ying Wu**, Gurumurthy Swaminathan, Zhizhong Li, Avinash Ravichandran, Nuno Vasconcelos, Rahul Bhotika and Stefano Soatto, “Class-Incremental Learning with Strong Pre-trained Models”, In CVPR, 2022. *[Paper]*
- Alakh Desai\*, **Tz-Ying Wu\***, Subarna Tripathi, and Nuno Vasconcelos, “Learning of Visual Relations: The Devil is in the Tails”, In ICCV, 2021. *[Paper]*
- **Tz-Ying Wu**, Pedro Morgado, Pei Wang, Chih-Hui Ho, and Nuno Vasconcelos, “Solving Long-tailed Recognition with Deep Realistic Taxonomic Classifier”, In ECCV, 2020. *[Paper]*

## SKILLS

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**Programming:** Python, C/C++, MATLAB, Verilog, Assembly, HTML, L<sup>A</sup>T<sub>E</sub>X

**Library:** PyTorch, Tensorflow, HuggingFace, vLLM, SGLang, Gradio, OpenCV, Pandas, Matplotlib

**System & Tool:** Linux, Git, Kubernetes, AWS, Azure, Docker